

Course Activities - Fundamentals of Semiconductor Device Physics and Process Integration

Course code: TGS-2024049339

Aligned deck: Trainer Slides v7.1

Each activity has its own self-contained folder with a procedure, scenario, worksheet, evidence checklist and same-folder printable PDFs.

1. [Device Physics Evidence Map](#) - K1 | A1 - 35 minutes - concepts 18-104 - overview slide 423
2. [Process Module Architecture](#) - A1 - 40 minutes - concepts 119-248 - overview slide 424
3. [CMOS Manufacturing Flow and Control Gates](#) - A2 - 50 minutes - concepts 249-287 - overview slide 425
4. [Lithography-Etch Interaction Case](#) - A3 - 45 minutes - concepts 138-209 - overview slide 426
5. [Doping and Thermal-Budget Interaction](#) - A3 - 45 minutes - concepts 210-228 - overview slide 427
6. [Yield and Performance Metric Verification](#) - K2 | A4 - 55 minutes - concepts 305-362 - overview slide 428
7. [Reliability Failure Analysis](#) - K2 | A4 - 45 minutes - concepts 363-373 - overview slide 429
8. [Follow-Up Action and Integration Review](#) - A5 - 60 minutes - concepts 407-422 - overview slide 430

Assessment alignment

- Activities 2-3 prepare evidence for Case Study Task 1 (A1-A2).
- Activities 4-8 prepare evidence for Case Study Task 2 (A3-A5).
- Activity 1 reinforces K1 device-physics reasoning used in the Written Assessment.
- Activities 6-7 reinforce K2 performance metrics used in the Written Assessment.